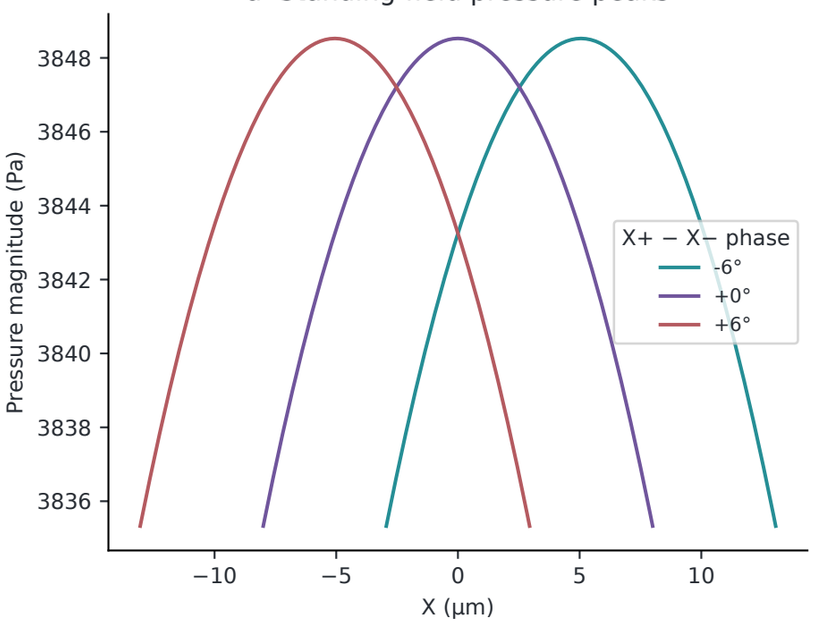
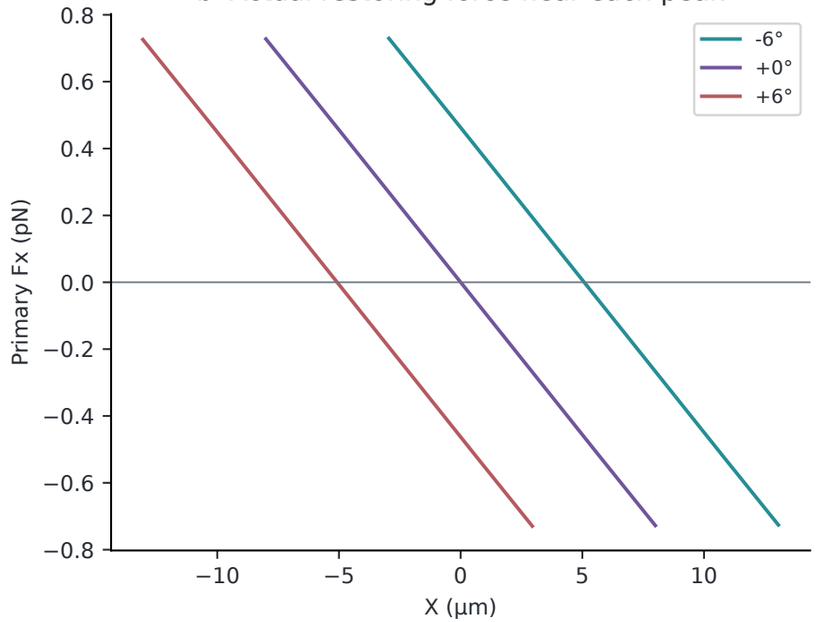
Phase-shifted pressure peaks: force stability and following limits

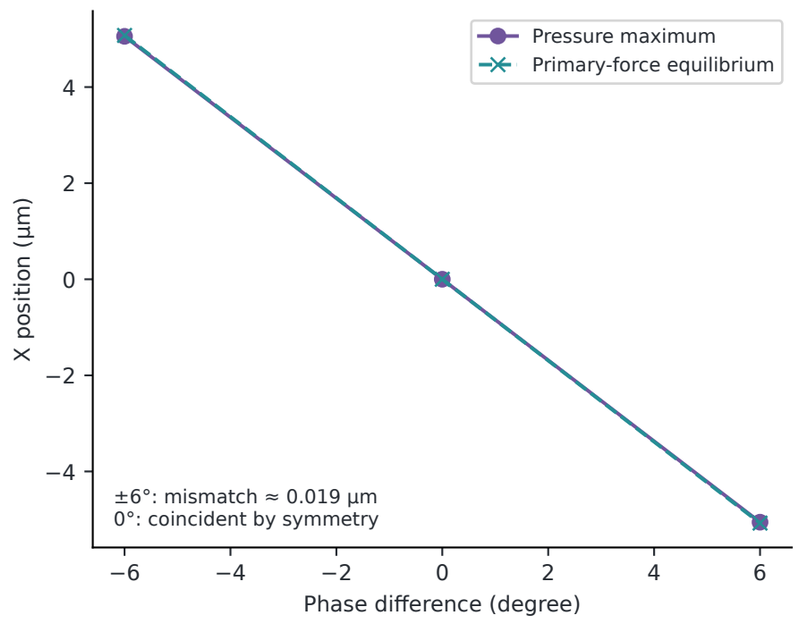
a Standing-field pressure peaks



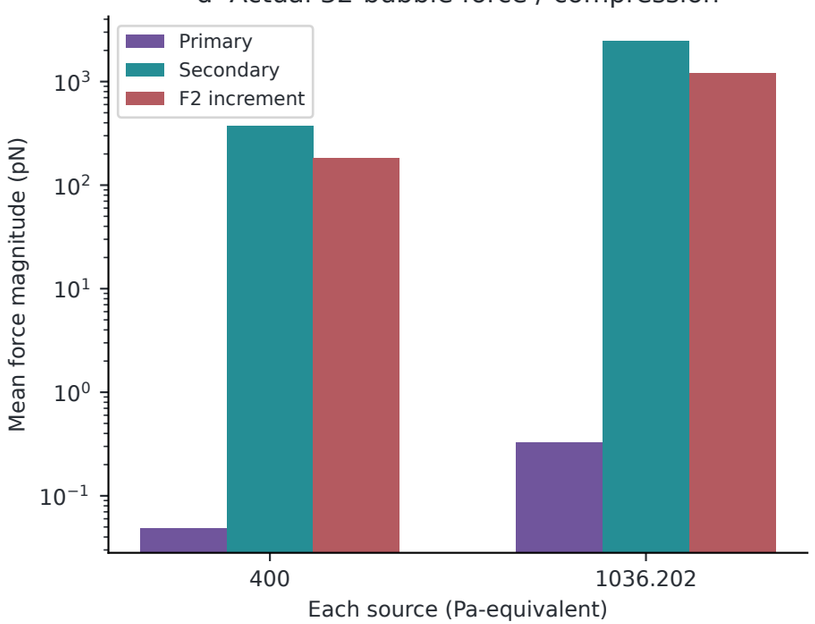
b Actual restoring force near each peak



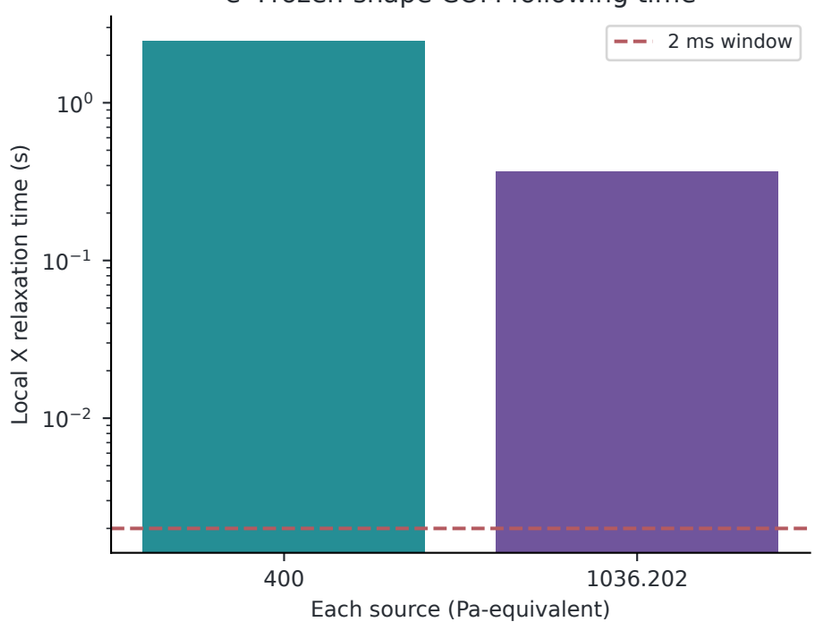
c Peaks and force zeros differ



d Actual 32-bubble force / compression



e Frozen-shape COM following time



f Stability does not imply fast following

LOCAL 3D RESTORING (proxy only)
All three force-Jacobian eigenvalues < 0

Frozen-shape diagnostic, NOT a trajectory
At 1036.202 Pa-equivalent:
  X relaxation: 0.368 s
  1° peak step: −0.843 μm
  Predicted 2 ms following: -0.0046 μm

Compression exists; no full trap contact admission
Long moving-trap trajectories: NOT_RUN
Rigid internal shape is not experimentally validated

APECSS and F2 evaluated at actual pressure · FROZEN_SHAPE_COM_DIAGNOSTIC · No long-time particle navigation claimed
EXPLORATORY · HOMOGENEOUS_WATER_CONTROL_PROXY · PHYSICS_UNCALIBRATED · training_ready=false